DE–7210 **11**

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION,
DECEMBER 2013.

ALGEBRA

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) If H and K are finite subgroups of G of orders O (H) and O(k) respectively then prove that O $(HK) = \frac{O(H)O(K)}{O(H \cap K)}$.

   (b) If ϕ is a homomorphism of G into $\overline{G}$ with kernel K, then prove that K is a normal subgroup of G.

2. (a) Prove that every permutation is the product of its cycles.

   (b) If P is a prime number and $P^{\alpha} | O(G)$, then prove that G has a subgroup of order p^{α}.

3. (a) Let R be a commutative ring with unit element whose only ideals are (O) and R itself. Then prove that R is a field.

   (b) Let R be a Euclidean ring. Then prove that any two elements a and b in R have a greatest common divisord. Moreover $d = \lambda a + \mu b$ for some $\lambda, \mu \in R$.

4. (a) If p is a prime number of the form 4n+1, then prove that $p = a^2 + b^2$ for some integers a, b.

(b) State the prove unique factorization theorem.

5. (a) If $v_1, \ldots\ldots v_n$ is a basis of V over F and if $w_1, \ldots\ldots w_m$ in V are linearly independent over F, then prove $m \le n$.

(b) Prove that two nilpotent linear transformations are similar if and only if they have same invariants.

6. (a) If F is of characteristic O and if S and T in $A_F(V)$, are such that ST–TS commutes with S, then prove that ST–TS is nilpotent.

(b) If $T \in A(V)$, then prove that $T^* \in A(V)$. More over prove the following.

(i) $(T^*)^* = T$

(ii) $(S+T)^* = S^* + T^*$

(iii) $(\lambda S)^* = \lambda S^*$

(iv) $(ST)^* = T^* S^*$ for all S,T,$\in A(V)$ and all $\lambda \in F$.

7. (a) Prove that the polynomial $f(x) \in F(x)$ has a multiple root if and only if $f(x)$ and $f'(x)$ have a nontrivial common factor.

(b) If $p(x)$ is a polynomial in $F(x)$ of degree $n \ge 1$ and is irreducible over F, then prove that there is an extension E of F, such that $[E:F] = n$,in which $p(x)$ has a root.

8. (a) If K is a finite extension of F, then prove that $G(K,F)$ is a finite group and its $O(G(K,F))$ satisfies $O(G(K,F)) \leq [K:F]$.

(b) Prove that for every prime number p and every positive integer m there exists a field having p^m elements.

DE–7211 **12**

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION, DECEMBER 2013.

REAL ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

$(5 \times 20 = 100)$

1. State and prove Bolzano-Weierstrass theorem. (20)

2. (a) Let f be a continuous mapping of a compact metric space X into a metric space Y. Then prove that f is uniformly continuous on X. (10)

   (b) State and prove the fixed point theorem for contraction mapping. (10)

3. (a) Let f be defined on $[a,b]$. If f has a local maximum at a point $x \in (a,b)$ and if $f'(x)$ exists then prove that $f'(x) = 0$. (10)

   (b) Assume that f' exists and is monotonic on an open interval (a,b). Prove that f' is continuous in (a,b). (10)

4. (a) Suppose f is a real differentiable function on $[a,b]$ and suppose $f'(a) < \lambda < f'(b)$. Then prove that there is a point $x \in (a,b)$ such that $f'(x) = \lambda$. (10)

(b) Suppose f is continuous on $[a,b]$. $f'(x)$ exists at some point $x \in (a,b)$. g is defined on an interval I which contains the range of f and g is differentiable at the point $f(x)$. If $h(t) = g(f(t))$ $(a \le t \le b)$ then prove that h is differentiable at x and $h'(x) = g'(f(x)f'(x)$. (10)

5. (a) If P^* is a refinement of P then prove that $L(p,f,\alpha) \le L(p^*,f,\alpha)$ and $U(p^*,f,\alpha) \le U(p,f,\alpha)$. (10)

(b) Prove that $\int_a^{-b} f d\alpha = \int_a^{-b} f d\alpha$. (10)

6. Suppose $\{f_n\}$ is a sequence of functions, differentiable on $[a,b]$ and such that $\{f_n(x_0)\}$ converges for some point x_0 on $[a,b]$. If $\{f_n'\}$ converges uniformly on $[a,b]$, then prove that $\{f_n\}$ converges uniformly on $[a,b]$ to a function f and $f'(x) = \lim_{n\to\infty} f_n'(x)$ $(a \le x \le b)$. (20)

7. (a) Prove that the interval (a,∞) is measurable. (10)

(b) Let $\langle E_n \rangle$ be an infinite decreasing sequence of measurable sets, that is sequence with $E_{n+1} C E_n$ for each n. Let mE_1 be finite. Then prove that $m\left(\bigcap_{i=1}^{\infty} E_i\right) = \lim_{n\to\infty} mE_n$. (10)

8. (a) State and prove Monotone convergence theorem. (10)

(b) Let f be a non negative function which is integrable over a set E. Then prove that given $\in > 0$ there is a $f > 0$ such that for every set $A \subset E$ with $mA < f$, we have $\int_A f < \in$. (10)

DE–7212 **13**

DISTANCE EDUCATION

M.Sc.(Mathematics) DEGREE EXAMINATION, DECEMBER 2013.

DIFFERENTIAL EQUATIONS AND NUMERICAL METHODS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries equal marks.

$(5 \times 20 = 100)$

1. Find y_2 and the general solution of each of the following equation.

   (a) $y''-y = 0,\ y_1(x) = e^x$. (10)

   (b) $xy''+3y' = 0,\ y_1 = 1$. (10)

2. (a) Derive Legendre equation in details. (10)

   (b) If the dependent variable is changed from y to ω by means of $y = \omega \cdot e^{\frac{-x^2}{4}}$, then show that the equation is transformed into $\omega - x\omega + p\omega = 0$. (10)

3. (a) Prove : If $y_1(x)$ and $y_2(x)$ are two linearly independent solutions of $y''+P(x)y'+Q(x)y = 0$, then the zeros of these functions are distinct and occur alternately – in the sense that $y_1(x)$ vanishes exactly once between any two successive zeros of $y_2(x)$ and conversely. (10)

(b) Find the normal form of Bessel's equation $x^2y''+xy'+(x^2-p^2)y=0$ and use it to show that every nontrivial solution has infinitely many positive zero. (10)

4. (a) Discuss about Charpit's method to solve PDEs. (10)

(b) Solve $pxy+pq+qy-yz=0$ using Charpit's method. (10)

5. (a) Let $q(x)$ be a positive continuous function that satisfies the inqualities $0<m^2<q(x)<m^2$ on a closed interval $[a,b]$. If $y(x)$ is nontrivial solutions $y''+q(x)y=0$, on this interval, and if x_1 and x_2 are successive zeros of $y(x)$, then $\frac{\pi}{M}<x_2-x_1<\frac{\pi}{m}$, furthermore, if $y(x)$ vanishes at a and b, and at $n-1$ points in the open interval (a,b), then $\frac{m(b-a)}{\pi}<n<\frac{M(b-a)}{\pi}$. (10)

(b) Prove : Let $q(x)$ be a positive continuous function and consider the differential equation $y''+\lambda\, q(x)y=0$ on a closed interval $[a,b]$. For each λ, let $y_\lambda(x)$ be the unique solution of the equation which satisfies the initial conditions $y_\lambda(a)=0$ and $y'_\lambda(a)=1$. Then there exists an increasing sequence of positive number $\lambda_1<\lambda_2<\cdots\lambda_n<\cdots$ that approaches ∞ and has the property $y_\lambda(b)=0$ if and only if λ equals one of the λ_n's. (10)

6. (a) Explain boundary value problem. (10)

(b) Find a non trivial solution y of the differential equation $y''+\lambda y=0$, subject to the conditions $y(0)=0$ and $y(\pi)=0$. (10)

7. Evaluate $\int_0^1 \frac{dx}{1+x^2}$ by using

(a) Trapezoidal rule. (10)

(b) Romberg's method, correct to four decimal places, Hence reduce an approximate value of π. (10)

8. Find the solution of differential equation $\frac{dy}{dx} = x^2 + y^2$ given $y(1) = 2.3$ using

(a) Euler's method. (10)

(b) Runge-Kutta method. (10)

DE–7213 **14**

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION, DECEMBER 2013.

OPERATIONS RESEARCH

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks. $(5 \times 20 = 100)$

1. (a) Solve the following LPP by graphically

Minimize $z = 2x_1 + x_2$

Subject to

$3x_1 + x_2 \geq 3$
$4x_1 + 3x_2 \geq 6$
$x_1 + 2x_2 \geq 2$ and

$x_1, x_2 \geq 0$.

(b) Solve the following LPs by the revised simplex method

Minimize $z = 5x_1 - 4x_2 + 6x_3 + 8x_4$

Subject to

$x_1 + 7x_2 + 3x_3 + 7x_4 \leq 46$
$3x_1 - x_2 + x_3 + 2x_4 \leq 20$
$2x_1 + 3x_2 - x_3 + x_4 \geq 18$
$x_1, x_2, x_3, x_4 \geq 0$.

2. (a) Determine the feasible solution to the following set of inequalities by using the dual problem

$2x_1 + 3x_2 \leq 12$

$-3x_1 + 2x_2 \leq -4$

$3x_1 - 5x_2 \leq 2.$

x_1, unrestricted and $x_2 \geq 0$.

(b) Consider the following LP :

Maximize $z = 2x_1 + 4x_2 + 4x_3 - 3x_4$

Subject to

$x_1 + x_2 + x_3 = 4$

$x_1 + 4x_2 + x_4 = 8$ and

$x_1, x_2, x_3, x_4 \geq 0.$

Use the dual problem to verify that the basic solution (x_1, x_2) is not optimal.

3. (a) Solve the following LPP using dynamic programming principles :

Maximize $z = 2x_1 + 5x_2$

Subject to

$2x_1 + x_2 \leq 43$

$2x_2 \leq 46$ and

$x_1, x_2 \geq 0.$

(b) Determine the saddle-point solution, the associated pure strategies and the value of the game for each of the of the following games. The payoffs are for player A.

	B_1	B_2	B_3	B_4
A_1	4	–4	–5	6
A_2	–3	–4	–9	–2
A_3	6	7	–8	–9
A_4	7	3	–9	5

4. (a) The following game give A's payoff. Determine the values of p and q that will make the entry (2, 2) of each game a saddle point.

	B_1	B_2	B_3
A_1	1	q	6
A_2	p	5	10
A_3	6	2	3

(b) Solve the following 2×4 game graphically. The payoff is for player A.

	B_1	B_2	B_3	B_4
A_1	2	2	3	–1
A_2	4	3	2	6

5. (a) Construct the project network comprised of activities A to L with the following precedence relationships.

(i) A, B and C, the first activities of the project, can be executed concurrently

(ii) A and B precede D

(iii) B precede E, F and H

(iv) F and C precede G

(v) E and H precede I and J

(vi) C, D, F and J precede K

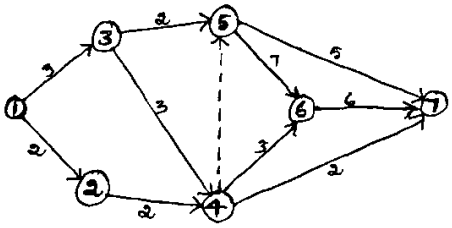
(vii) K precedes L

(viii) I, G and L are the terminal activities of the project.

(b) Explain critical path computations.

6. (a) A manufacturer has to supply his customer with 600 units of his product per year, shortages are not allowed and storage cost amounts to Re. 0.60/unit per year. The set-up cost per run is Rs. 80,000. Find the optimum run size and the minimum average yearly cost.

(b) Determine the critical path for the following project network.



7. (a) The time between failures of a kencore refrigerator is known to be exponential with mean value 9000 hours (about 1 year of operation) and the company issues a 1-year warranty on the refrigerator. What are the chances that a breakdown repair will be covered by the warranty?

(b) Explain multi-server models.

8. (a) For the (M/M/1) : (GD/∞/∞), derive the expression for L_p using the basic definition $\sum_{n=2}^{\infty}(n-1)P_n$.

(b) Customers arrive at thrift Bank according to a Poisson distribution, with a mean of 45 customers per hour. Transactions per customer last about 5 minutes and is exponentially distributed. The bank wants to use a single-line multiple-teller operation, similar to the ones used in airports and post offices. The manager is conscious of the fact that customers may switch to other banks if they perceive that their wait in line is "excessive". For this reason, the manager wants to limit the average waiting time in the queue to no more than 30 seconds. How many tellers should the bank provide?

DE–7214 **15**

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION,
DECEMBER 2013.

MATHEMATICAL STATISTICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

$(5 \times 20 = 100)$

All questions carry equal marks.

1. (a) If X and Y are independent Gamma variates with parameters μ and v respectively, show that the variables $U = X + Y$, $Z = \dfrac{X}{X+Y}$ are independent and that U is a $\delta(\mu + v)$ variate and z is a $\beta_1\ (\mu, v)$ variate.

(b) The joint probability distribution of two random variables X and Y is given by: $P(X = 0, Y = 1) = \dfrac{1}{3}$, $P(X = 1, Y = -1) = \dfrac{1}{3}$ and $P(X = 1, Y = 1) = \dfrac{1}{3}$. Find

(i) Marginal distributions of X and Y and

(ii) Conditional probability distribution of X given $Y = 1$.

2. (a) Given the joint density function of X and Y as

$$f(x, y) = \begin{cases} \frac{1}{2} xe^{-y}; & 0 < x < 2 \ , y > 0 \\ 0 & otherwise \end{cases}$$

Find the distribution of $X + Y$.

(b) Let $X \sim \beta_1(\mu, v)$ and $Y \sim \delta(\lambda, \mu + v)$ be independent random variables, $(\mu, v, \lambda > 0)$. Find a p.d.f for XY and identify its distribution.

3. (a) (i) Let X_i assume that values i and $-i$ with equal probabilities. Show that the law of large numbers cannot be applied to the independent variable $X_1, X_2.....(ie), X_i$ is

(ii) If X_i can have only two values with equal probabilities i^α and $-i^\alpha$, show that law of large numbers can be applied to the independent variables $X_1, X_2,.....$, if $\alpha < \frac{1}{2}$.

(b) Find the maximum likelihood estimate for the parameter λ of a Poisson distribution.

4. (a) The height of six randomly chosen sailors are (in inches) : 63, 65, 68, 69, 71 and 72. Those of 10 randomly chosen soldiers are 61, 62, 65, 66, 69, 69, 70, 71, 72 and 73. Discuss the light that these data throw on the suggestion that sailors are on the average taller than soldiers.

(b) A certain stimulus administered to each of the 12 patients resulted in the following increase of blood pressure : 5, 2, 8, –1, 3, 0, –2, 1, 5, 0, 4 and 6.

Can it be concluded that the stimulus will, in general, be accompanied by an increase in blood pressure?

5. (a) Write down the procedure for testing of hypothesis for large samples.

   (b) In a big city 325 men out of 600 men were found to be smokers. Does this information support the conclusion that the majority of men in this city are smokers?

6. (a) The means of two samples of 1000 and 2000 members are respectively. 67.5 and 68. Can they be regarded as drawn from the same population with S.D. 2.5?

   (b) Explain test of significance for difference of proportions.

7. (a) Write down the uses of S.Q.C.

   (b) The following are the figures of defectives in 22 lots each containing 2,000 rubber belts.

   425, 430, 216, 341, 225, 322, 280, 306, 337, 305, 356, 402, 216, 264, 126, 409, 193, 326, 280, 389, 451, 420.

   Draw control chart for fraction defective and comment on the state of control of the process.

8. (a) A test was given to five students taken at random from the fifth class of three schools of a town. The individual scores are

School I :	9	7	6	5	8
School II :	7	4	5	4	5
School III :	6	5	6	7	6

   carry out the analysis of variance, and state your conclusions.

   (b) Work out the analysis of variance for a two way classification assuming that intersection is present. State the assumptions made.

DE–7215 **21**

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION,

DECEMBER 2013.

COMPLEX ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.
All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Prove that two points z_1 and z_2 are reflection points for the line $\bar{\alpha}z + \alpha\bar{z} + \beta = 0$ iff $\bar{\alpha}z_1 + \alpha\bar{z}_2 + \beta = 0$.

   (b) State and prove Cauchy-Riemann equations.

2. (a) Prove that any bilinear transformation which maps the real axis onto unit circle $|w| = 1$ can be written in the form $w = e^{i\lambda}\left(\dfrac{z-\alpha}{z-\bar{\alpha}}\right)$ where λ is real. Further this transformation maps the upper half plane $\operatorname{Im} z \geq 0$ onto the unit circular disc $|w| \leq 1$ iff $\operatorname{Im} \alpha > 0$.

   (b) Discuss the mapping $w = e^z$.

3. (a) State and prove maximum modulus theorem.

   (b) Evaluate $\displaystyle\int_C \frac{e^z}{z^2+4}\,dz$ where C is positively oriented circle $|z - i| = 2$.

   (c) Evaluate $\displaystyle\int_C \frac{\sin z}{\left(z - \frac{\pi}{2}\right)^2}\,dz$ where C is the circle $|z| = 2$.

4. (a) State and prove Taylor's theorem.

   (b) State and prove Weiestrass theorem on essential singularity.

5. (a) Find the Laurent's series expansion of the function $\frac{z^2 - 1}{(z+2)(z+3)}$ valid in the annular region $2 < |z| < 3$.

   (b) State and prove Cauchy's residue theorem.

6. (a) Prove that every function which is meromorphic in the whole plane is the quotient of two entire functions.

   (b) State and prove Hadamard's theorem.

7. (a) Prove that any two bases of the same module are connected by a unimodular transformation.

   (b) Prove that

$$\begin{vmatrix} p(z) & p'(z) & 1 \\ p(u) & p'(u) & 1 \\ p(u+z) & -p'(u+z) & 1 \end{vmatrix} = 0 .$$

8. (a) Show that any elliptic function with periods w_1, w_2 can be written as $C\prod_{k=1}^{n} \frac{\sigma(z - a_k)}{\sigma(z - b_k)}$ (c = constant).

   (b) Prove that a non constant elliptic function has equally many poles as it has zeros.

DE–7216 **22**

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION,
DECEMBER 2013.

TOPOLOGY AND FUNCTIONAL ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. Prove that every separable metric space is second countable.

2. (a) Prove that any closed subspace of a compact space is compact.

   (b) Prove that any continuous image of a compact space is compact.

3. (a) Prove that a topological space is a T_1-space if and only if each point is a closed set.

   (b) Prove that every compact Housdorff space is normal.

4. (a) State and prove Uryson's Lemma.

   (b) Prove that any continuous image of a connected space is connected.

5. (a) State and prove the closed Graph theorem.

   (b) State and prove the uniform Boundedness theorem.

6. (a) Prove that if x and y are any two vectors in a Hilbert space then $|(x, y)| \leq \|x\| \ \|y\|$.

(b) State and prove Schwarz inequality.

7. (a) If $\{e_i\}$ is an orthonormal set in a Hilbert space H then prove that $\Sigma|x, e_i|^2 \leq \|x\|^2$ for every vector x in H.

(b) Prove that the adjoint operation $T \to T^*$ on $\mathbf{B}(H)$ has the following properties.

(i) $(T_1, T_2)^* = T_2^* \ T_1^*$

(ii) $\|T^* \ T\| = \|T\|^2$.

8. (a) If T is an operator on H for which $(Tx, x) = 0$ for all x then prove that $T = 0$.

(b) Prove that an operator T on H is unitary $\Leftrightarrow$ it is an isometric isomorphism of H onto itself.

DE–7217 **23**

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION,
DECEMBER 2013.

GRAPH THEORY

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Define walk and prove that a closed walk of odd length contains a cycle.

   (b) Let G be a (p, q) graph. Prove that the following statements are equivalent.

   (i) G is a tree

   (ii) Every two points of G are joined by a unique path

   (iii) G is connected and $p = q + 1$

   (iv) G is acyclic and $p = q + 1$.

2. (a) Prove that any tournament on $2K$ vertices contains a copy of each branching on $K + 1$ vertices.

   (b) Show that an edge e of G is a cut edge of G if and only if e is contained in no cycle of G.

3. (a) Let G be a 3–connected graph on atleast five vertices. Then prove that G contains an edge e such that G/e is 3-connected.

   (b) State and prove Dirac theorem.

4. (a) Define closure of a graph G and prove that $C(G)$ is well defined.

   (b) Prove that a connected graph has an Euler trail if and only if it has atmost two vertices of odd degree.

5. (a) State and prove Gallai-Rai Theorem.

   (b) For any positive integer K, show that there exists a K-chromatic graph containing no triangle.

6. (a) For any positive integer K, prove that there exist a triangle-free K-chromatic graph.

   (b) State and prove the combinatorial Nullstellensatz.

7. (a) Prove that an Inner (outer) bridges avoid one another.

   (b) State and prove five colour theorem.

8. (a) Prove that the edges of a connected graph $G = (V, E)$ can be oriented so that the resulting digraph is strongly connected iff every edge of G is contained in atleast one cycle.

   (b) Prove that a weak digraph D is Eulerian iff every point of D has equal indegree and outdegree.

DE–7218 **24**

DISTANCE EDUCATION

M.Sc. (Mathematics) DEGREE EXAMINATION,
DECEMBER 2013.

PROGRAMMING IN C/C++

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain constants, variables and their declaration in 'C'. (10)

   (b) Explain data types and functions. (10)

2. (a) Explain the structure of a 'C' program. (10)

   (b) Explain keywords and literals. (10)

3. (a) What are the general forms of scanf(), printf(), gets(), gethar () functions? Explain. (10)

   (b) What do you mean by control codes and arithmetic operators. (10)

4. (a) Explain increment and assignment operators. (10)

   (b) What do you understand about the local and global variables? (10)

5. (a) Explain for loop with suitable program in C++. (10)

   (b) Write a C++ program to find the sum of the 5 digits of the number. (10)

6. (a) How to declare arrays and strings? Explain. (10)

   (b) Explain

      (i) Comparing two strings. (5)

      (ii) Determining string length. (5)

7. (a) (i) Define printers. (4)

      (ii) How to assigning structure variables and initial values? (6)

   (b) Explain: Inline functions in C++. (10)

8. (a) Explain: filebuf, if stream, offstream fstream. (10)

   (b) Explain: write () and read() functions with example. (10)

DE–7219 **25**

DISTANCE EDUCATION

M.Sc. DEGREE EXAMINATION, DECEMBER 2013

Mathematics

DISCRETE AND COMBINATORIAL MATHEMATICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) If $a_n = Aa_{n-1} + Ba_{n-2,}$ and if $S_n = a_1 + a_2 + ... + a_n$, prove that $S_{n+3} = (A+1)S_{n+2} + (B-A)S_{n+1} - BS_n$. Verify this in the case A=B=1, where the a_n are the Fibonacci numbers. (10)

   (b) Find a formula for the general term F_n of the Fibonacci sequence 0,1,1,2,3,5,8,13, ……. (10)

2. (a) Prove that the elements f of $\mathbf{R}[t]$ of given by $f = \sum_{k=0}^{\infty} \alpha_k t^k$ has an inverse $\mathbf{R}[t]$ iff α_0 has an inverse in $\mathbf{R}$. (10)

   (b) If you satisfies $y_{n+1} - 2y_n Cos\alpha + y_{n-1} = 0$ for n=1,2 …. and if n=1,2, ….. and if $y_0 = 0, y_1 = 1$ find y_2, y_3, y_4. (10)

3. (a) Prove that $\varepsilon(t) = \sum_{j=0}^{N} W(j)(t-1)^j$. (10)

(b) How many solutions does the equation $x_1 + x_2 + x_3 = 11$ have where x_1, x_2, x_3 are non negative such that $x_1 \le 3, x_2 \le 4$ and $x_3 \le 6$? Use the principle of inclusion-exclusion. (10)

4. (a) Let a deck of n distinct cards be matched with another deck of n cards, both well shuffled. In a random matching of the two, card by card, what is the number of ways of having exactly m matches? (10)

(b) In how many ways can the integers 1 through 9 be permuted such that

(i) No odd integer will be in its natural position?

(ii) No even integer will be in its natural position? (5+5)

5. (a) Define G-equivalence and prove that G-equivalence, is an equivalence relation on the set R^D of all functions from D to $\mathbf{R}$. (10)

(b) State and Prove Polya's fundamental theorem. (10)

6. (a) State and Prove Polya's enumeration theorem. (10)

(b) Prove that : Let $(\mu) = 1^{\mu_1} 2^{\mu_2} .....n$. $M_{\lambda\mu}$ is the number of functions $\phi : D \to R$, D and R both n-sets, which have the properties $W(\phi) = \alpha_1^{\mu_1} \alpha_2^{\mu_2} .... = w, g\phi = \phi(g \in S_n \text{ of type} \lambda)$. (10)

7. (a) Let G be a group. Let L be the set of all subgroups of G. In L we define $A \leq B$ iff $A \leq B$. Then prove that L is a lattice. (10)

(b) Define switching circuit and switching function. Explain them through an example. (10)

8. (a) Let B be Boolean algebra. Then prove that

(i) $(a \vee b)' = a' \wedge b'$ and $a \wedge b)' = a' \vee b'$. (3+3)

(ii) $(a')' = a$. (2)

(b) In a Boolean algebra if $a \vee x = b \vee x$ and $a \vee x' = b \vee x'$, then prove that $a = b$. (4)

(c) (i) Show that in a Boolean algebra B, the complement of any element is not itself. (4)

(ii) Prove that a Boolean algebra cannot have exactly three elements. (4)